Smart Donation Connect: An AI-Enabled Platform for Reducing Waste and Connecting Donors with People in Need

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ABSTRACT: Resource wastage and uneven distribution are significant social and environmental issues. While hotels, restaurants, businesses, and households may have an excess of food, clothing, electronic items, and other resources, NGOs, orphanages, old-age homes, and economically backward areas may experience shortages. As a result, Smart Donation Connect is an AI-based web application designed to connect verified donors with verified recipients of donations. The application offers secure user registration and verification, donation posting, beneficiary requirement posting, smart donor-to-beneficiary matching, location-based search and matching, a notification system, and donation tracking. The AI-based food demand prediction system uses historical sales data, seasonal variations, festive seasons, weather systems, and customer behavior to predict food demand and forecast surplus. The AI-based recommendation system helps match the right beneficiary with the right kind of donation, location, urgency, quantity, and requirements. The application tracks donations from posting, matching, and accepting to collecting, delivering, and completing the donation process. Smart Donation Connect aims to solve resource wastage, increase charitable organizations' agility, and ensure that resources reach genuine recipients in need while promoting community goodwill.

Keywords: Artificial Intelligence, Smart Donation, Food Waste Prediction, Donation Matching, Resource Sharing, Location-Based Matching, Social Welfare, Real-Time Notifications, Sustainable Development.

I. INTRODUCTION

Large amounts of food, clothing, electronic devices, educational materials, and household resources are wasted even though many people and organizations require these resources. Restaurants, hotels, educational institutions, households, and businesses may have surplus resources that could help others. However, without an organized platform, it is often difficult to identify genuine recipients and distribute available resources efficiently.

Traditional donation systems generally depend on manual searching, personal contacts, or social media communication. These approaches may not efficiently identify suitable recipients based on their actual needs. In addition, donors may not have sufficient information about the quantity, urgency, location, and condition of the resources required by receivers.

To address these challenges, this paper proposes Smart Donation Connect, an AI-enabled platform that connects donors with verified receivers. The system uses intelligent matching techniques to consider resource type, quantity, condition, urgency, and geographical location. In addition, the platform can analyze historical food-related information to predict future demand and identify possible surplus resources.

The proposed system aims to reduce resource wastage while improving the efficiency and reliability of the donation process. It provides a structured platform through which donors can list available resources and receivers can post their requirements. The system then assists in identifying suitable donor–receiver pairs and tracks the donation process until completion.

II. RELATED WORK

Several existing approaches have investigated the use of technology and machine learning for food donation, resource sharing, and waste reduction. Machine-learning-based food demand forecasting has been used to estimate future requirements and improve resource planning. Such forecasting techniques can help organizations identify periods of high or low demand.

Food-sharing platforms have also been developed to connect organizations or individuals having surplus food with people who can utilize it. However, many existing systems primarily focus on food sharing and may not support multiple categories of resources such as clothing, electronics, educational materials, and household items.

Location-based resource-sharing systems can improve the distribution process by connecting users who are geographically closer to each other. However, location information alone may not be sufficient because the quantity, condition, urgency, and category of a resource must also be considered.

The proposed system combines these concepts into a single platform by integrating resource listing, requirement posting, demand prediction, intelligent matching, location-based matching, notifications, and donation tracking.

III. METHODOLOGY

The proposed Smart Donation Connect system is designed to create an AI-based connection between verified donors and genuine beneficiaries. The methodology consists of user registration and verification, donor and receiver modules, AI-based food demand prediction, intelligent donor-receiver matching, location-based matching, notifications, and donation tracking.

A. User Registration and Verification

The system offers separate registration for donors and beneficiaries. The system considers all types of donors, including individuals, hotels, restaurants, business owners, NGOs, and organizations. Similarly, beneficiaries include orphanages, old-age homes, educational institutions, and economically backward families. The basic user details and contact information are collected and verified to improve the reliability and authenticity of the registered users.

B. Donor and Receiver Modules

The Donor Module helps the user list the available resources, including food, clothes, electronics, educational material, home needs, and monetary donations. The user can enter more details about the available items. The Receiver Module helps verified users to enter their demands in terms of resources, quantity, urgency, and location.

C. AI-Based Food Demand Prediction

The system uses historical data of restaurants and hotels to predict future food demand and estimate surplus. The process considers previous sales, customer behavior, seasonal variations, festive demands, and weather conditions, among others. The process follows Data Collection → Preprocessing → Feature Selection → ML Model → Demand Prediction → Surplus Estimation.

D. Intelligent Donor-Receiver Matching

The Matching Module matches the resource list of donors with the needs of the beneficiaries. It considers the location of the beneficiary and the donor, category of donation, level of urgency, quantity demanded and

supplied, and other factors. The system recommends suitable donors and beneficiaries to ensure that the available resources reach the right customers.

E. Notifications, Location and Donation Tracking
The system offers notification alerts to both the donors and beneficiaries when a suitable match is found. Location-based matching helps find the nearest beneficiaries and available resources, especially for time-sensitive food donations. The system tracks the donation process through Posted → Matched → Accepted → Collected → Delivered → Completed stages.

The methodology uses AI prediction and matching, along with location tracking and donation alerts, to develop a reliable donation management system that minimizes wastage and increases donation efficiency.

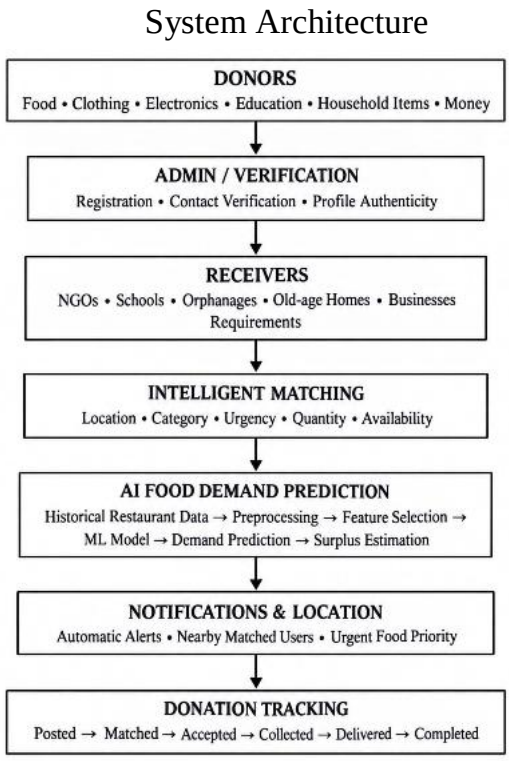


Fig. 1. System Architecture of Smart Donation Connect

The system architecture below represents the flow of a donation within the system, starting from the registration of a donor until a successful donation is completed.

1. DONORS
This segment deals with potential donors with available resources to donate. These resources include:
Food
Clothes
Electronics
Educational materials
Household goods
Money
Other information about the donated resources, such as quantity and condition and their availability, location can be specified.

2. ADMIN / VERIFICATION
The Admin/Verification segment deals with verifying the legitimacy of the users. This includes registration, contact information verification, and profile verification. This step ensures that all donors and receivers are legitimate and therefore enhances the trust and reliability of the platform among its users.

3. RECEIVERS
This segment is for the receivers of the donated resources. These receivers range from NGO's, schools, orphanages, old-age homes, and businesses. Details about the specific requirements and quantities of the resources needed can be specified and saved in this segment.

4. INTELLIGENT MATCHING
The intelligent matching segment deals with finding a suitable match for a donor based on location, category, urgency, quantity, and availability. For instance, in the case of food donation, a needy receiver identified through the verification process closer to the location of the donor can be recommended.

5. AI FOOD DEMAND PREDICTION
This segment is an AI/ML segment that deals with predicting demand for food. For restaurants and hotels, historical data is utilized in this system to predict future demand for food. The following steps

are involved in this process: For instance, food surplus estimation is crucial for the donation process.

6. NOTIFICATIONS & LOCATION

The notifications segment deals with notifying the users about the progress of their donations.
It can also help to recommend location-based recipients and prioritize urgent food donations.

7. DONATION TRACKING

The donation tracking segment monitors the status of a donation from being posted until it is marked as a completed donation. This segment collects information about the donor, receiver, resources donated, number of resources, date, and status of the donation.

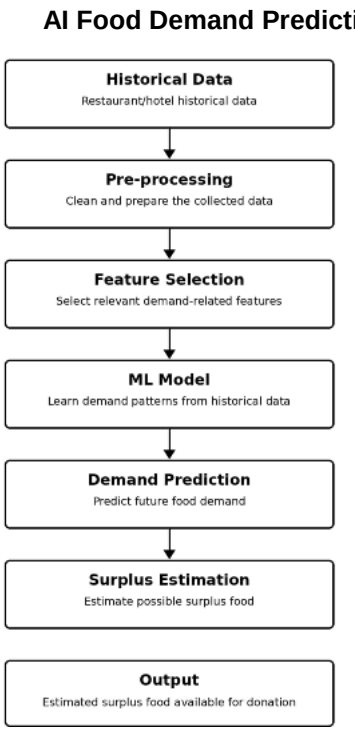


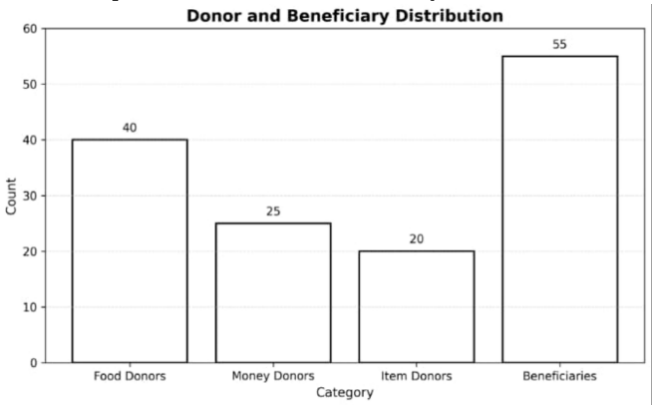
Fig. 2. AI Food Demand Prediction Process

The system includes Donor, Receiver, and Admin modules. Donors provide available resources, while receivers post their requirements. The

Admin manages registration and verification. The Matching Module connects suitable donors and receivers based on location, urgency, and quantity. The AI Demand Prediction Module estimates food demand and potential surplus using historical data. Notifications inform users about suitable matches, while the Tracking Module monitors the donation process until completion.

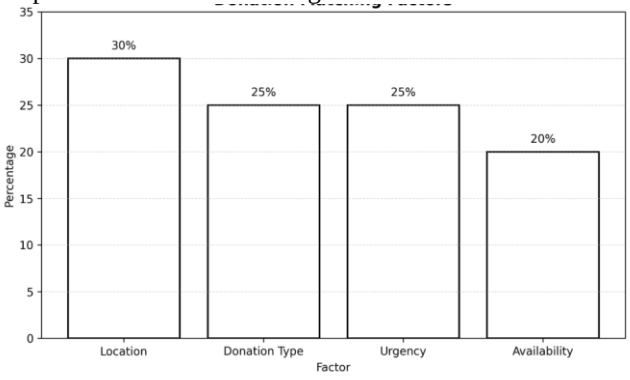
Fig. 2 illustrates the AI-based food demand prediction process. Historical restaurant data is first collected and preprocessed. Relevant features are then selected and provided to the machine learning model. The model predicts future food demand, which is used to estimate potential surplus food.

Graph 1 — Donor and Beneficiary Distribution



Graph 1 represents the participation of different user categories in the proposed donation platform. Beneficiaries form the largest group, while food donors represent a significant portion of the donor community. The system is designed to connect these users according to donation type, location, and requirement.

Graph 2— Donation Matching Factors



Explanation:

The proposed matching mechanism considers multiple factors when connecting donors with beneficiaries. Location

receives the highest proposed weight because nearby donations can be delivered more efficiently. Donation type and urgency are also considered to improve the relevance of each match.

IV. RESULTS & DISCUSSION

The proposed Smart Donation Connect is an innovative platform that deals with user registration and verification, managing donations, smart donor–receiver matching, AI-enabled food demand prediction, location-based matching, notifications, and tracking. It is a system designed to establish a connection between those who have surpluses and those in need of goods.

The first two modules, namely, the donor and receiver modules, enable users to enter information concerning the resources they want to donate or look for in terms of donations. The former ensures that the necessary data about an item for donation is posted, whereas the latter allows for specifying the required resource and its characteristics. The matching algorithm is responsible for finding relevant donor–receiver pairs on the basis of such criteria as the category of a resource, location, level of urgency, quantity, and availability.

The AI-based prediction of food demand and supply entails an analysis of historical and current data in order to determine the approximate level of food demand and supply and help predict food surpluses that could be donated. It could allow for an earlier identification of a surplus and a better distribution of goods based on such factors as the level of urgency of need and location. In addition, the modules for notifications and tracking ensure that users are informed about the progress of their donations and all changes in terms of posting, matching, accepting, collecting, and completing a donation.

The discussed system shows how a particular platform could leverage such technologies as AI-based prediction, smart matching, and location-based services to address the issue of donations and manage the process more efficiently. At the same time, there are concerns regarding the accuracy of the results, especially when it comes to the AI-based prediction module. To evaluate it, one would need to analyze actual data to see how it fits historical trends and determine whether the predicted values match the actual ones. Moreover, it is vital to conduct performance testing to see how well the system will perform under a specific data load and how fast it will process data and find matches.

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